

Double angle identities



1

a $\frac{\sqrt{3}}{2}$

b $\frac{\sqrt{2}}{2}$

c $\frac{\sqrt{2}}{2}$

d $\frac{\sqrt{3}}{2}$

e $\frac{5\sqrt{2}}{2}$

f $-\frac{\sqrt{2}}{24}$

g $\frac{3\sqrt{3}}{2}$

h $-\frac{\sqrt{2}}{8}$

2 $-\frac{5\sqrt{2}}{2}$

3

a $\sin \theta = -\frac{3}{5}$

b $-\frac{24}{25}$

c $\frac{7}{25}$

4

a $\cos \theta = -\frac{24}{25}$

b $\frac{336}{625}$

c $\frac{527}{625}$

5

a $\frac{24}{25}$

b $\frac{7}{25}$

6

a $\cos^2 \left(\frac{x}{2} \right)$

b $\cos x = \frac{5}{13}$

c $-\frac{3\sqrt{13}}{13}$

7

a $\cos^2 \left(\frac{x}{2} \right)$

b $-\frac{12}{13}$

c $\frac{\sqrt{26}}{26}$

8

a $\sin^2 \left(\frac{x}{2} \right)$

b $\frac{\sqrt{17}}{17}$

c $\frac{\sqrt{578 - 34\sqrt{17}}}{34}$

9

a $\frac{\sqrt{2 + \sqrt{2}}}{2}$

b $-\frac{\sqrt{2 - \sqrt{3}}}{2}$

c $\frac{\sqrt{2 + \sqrt{2}}}{2}$

d $-\frac{\sqrt{2 + \sqrt{2}}}{2}$

10

a $\frac{\sqrt{154}}{14}$

b $\cos \left(\frac{x}{2} \right) = \frac{\sqrt{70}}{10}$

c $-\frac{\sqrt{30}}{6}$

d $-\frac{\sqrt{154}}{14}$

11 q

12 a No solution provided.



b $\sin 18^\circ = \frac{4 \cos^2 (18^\circ) - 3}{2}$

13

a No solution provided.

b No solution provided.

c No solution provided.

d No solution provided.

e No solution provided.

f No solution provided.

Equations

14

a $x = \frac{\pi}{3}, \frac{5\pi}{3}, 0$

b $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$

c $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$

d $x = 128.2^\circ, 231.8^\circ$

e $x = 90^\circ, 270^\circ, 135^\circ, 315^\circ$

15

a No solution provided.

b $\theta = 0, \pi, \frac{\pi}{6}, \frac{5\pi}{6}$

16

a $1 - \sin 2\theta$

b The equation has at least 1 real solution.

Applications

17

a $\theta = 84$

b $m = 1.4$

18

a $\frac{755}{4} \sin^2 (110\pi t)$

b Maximum = $\frac{755}{4}$
Minimum = 0

c $-\frac{755}{8} \cos 220\pi t + \frac{755}{8}$

d 90 W

19

a $x = \cos a$

c No solution provided

e $\frac{\sqrt{2}}{2}$



b $y = \sin a$

d $a = 45^\circ$

f $\sqrt{2}$